Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6 HT	(a)			**Indicative content:** The National Grid consists of a network of power stations, transformers/sub-stations and power lines all joined together supplying electricity to consumers. Supply and demand is monitored. At times of power station failure electricity can be transferred from one part of the Grid to another. At times of peak demand, reserve generation can increase supply or electricity can be imported from abroad. Transformers are used to step-up the voltage before transmission along power lines. This reduces current and power loss as heat in the cables. **5-6 marks** Full description of the 3 aspects i.e. the National Grid and how it achieves reliability and greater efficiency. *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3 -4 marks** Full description of 2 out of the 3 aspects i.e. the National Grid and how it achieves reliability and greater efficiency **or** brief description of all 3 aspects. *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Full description of 1 out of the 3 aspects i.e. the National Grid or how it achieves reliability or greater efficiency **or** limited description of up to 2 aspects. *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*	6			6		

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		63 - 1.42 = 61.58 **or** 63 000 – 1 420 = 61 580 (1) Substitution: $\frac{61.58\ \textbf{ecf}}{63} \times 100$ **or** $\frac{61\ 580\ \textbf{ecf}}{63\ 000} \times 100$ (1) % efficiency = 97.7[5] (1) **Alternative (example of a wasted energy route):** $\frac{1.42}{63} = 0.0225$ (1) 0.0225 **ecf** × 100 = 2.25% (1) so % efficiency = 100 – 2.25 = 97.7[5] (1) **Alternative (max demand off the winter curve):** 55 000 – 1 420 = 53 580 (1) $\frac{53\ 580\ \textbf{ecf}}{55\ 000} \times 100$ (1) So % efficiency = 97.4 (1) **Alternative (max demand off the summer curve):** (43 000 ± 1 000) – 1 420 = 41 580 (1) $\frac{41580\ \textbf{ecf}}{43\ 000} \times 100$ (1) So % efficiency = 96.7 (1) **Alternative (calculating input energy)** Useful energy = 63 000 [MW] Input energy = 63 000 + 1 420 = 64 420 [MW] (1) $\frac{63\ 000}{64\ 420\ \textbf{ecf}} \times 100$ (1) So % efficiency = 97.8 (1) **N.B.1** If in any of the above methods 1 420 has not been subtracted then **award no marks**	 1	1 1		3	3	

(Marks Available: AO1, AO2, AO3, Total, Maths, Prac)

Question				Marking details	Marks Available AO1	AO2	AO3	Total	Maths	Prac
				N.B.2 63 000 – 1 420 = 61 580 (1) $\frac{61580}{79\,900}$ no further credit to be given and for $\frac{63\,000}{79\,900}$ no credit to be given **N.B.3** Attempt at subtracting 1 420 MW from 63 GW or 79.9 GW where the conversion is incorrect **award 1 mark only provided the outcome is positive** **N.B.4** Ignore rounding errors on the % value **Accept** - If output capacity used 79 900 – 1 420 = 78 480 (1) $\frac{78\,480\ \textbf{ecf}}{79\,900}$ x 100 = 98.2% (1) so **award a maximum of 2 marks**						
		(ii)		Substitution: $\frac{360\text{ TWh}}{8\,766\text{ h}}$ (1) Conversion i.e. $\frac{360\,000}{8\,766}$ (1) = 41[.1] [GW] (1)	1	1 1		3	3	
		(iii)		Ticks in boxes 1 and 3 (for 1 mark each) i.e. At maximum demand there is spare capacity of 16.9 GW (1) The difference in minimum demand between summer and winter is approximately 10 000 MW (1) Lose 1 mark for each extra tick			2	2		
				Question 6 total	**8**	**4**	**2**	**14**	**6**	**0**